A. Internationalization, Localization & Regional Behavior

Standalone capability brief · Suggested module: **Globalization SDK + Locale Service**

Purpose

Provide a consistent way for product teams to support languages, locales, regional rules, and region-specific deployments without reimplementing globalization logic in every domain.

1. Scope and capability areas

Internationalization (i18n)

Design APIs, data models, and interfaces so text expansion, multiple scripts, plural forms, locale-aware formatting, and translation can be supported without structural rewrites.

Localization (l10n)

Adapt product content, messages, terminology, and workflows for a particular language or market. Keep translated content separate from application logic.

Formatting

Standardize locale-aware dates, times, currencies, numbers, percentages, measurement units, and calendars. Avoid hand-built formatting and hard-coded symbols.

Time zones

Store event instants in a consistent representation, normally UTC, and render them in the user's or business's selected time zone. Handle daylight-saving transitions and ambiguous local times explicitly.

Right-to-left support

Support RTL layout direction, mirrored navigation where appropriate, bidirectional text, and mixed-direction content such as numbers or identifiers.

Translation management

Provide catalog versioning, fallback languages, pluralization, translation review workflows, missing-key detection, and safe deployment of translation updates.

Locale-aware search and sorting

Define Unicode normalization, collation, case handling, and language-specific search behavior. Avoid assuming ASCII or English sorting rules.

Regionalization

Represent data-residency regions, regional service endpoints, legal notices, market-specific business rules, and region-dependent availability explicitly.

2. Shared contracts and standards

- Resolve locale and time-zone preferences using a documented precedence order, such as user preference → tenant default → product default.

- Expose stable SDK helpers for formatting and message lookup; avoid domain services inventing their own locale conventions.
- Version translation catalogs and define behavior for missing keys, unavailable locales, and stale clients.
- Keep regulatory and business rules owned by the relevant domain, while providing reusable mechanisms for regional configuration.

3. Recommended implementation approach

- Create a locale context object containing language tag, formatting locale, time zone, direction, and region where applicable.
- Use standard locale identifiers and mature internationalization libraries rather than custom string manipulation.
- Store canonical timestamps and currency amounts with explicit currency codes; do not infer currency from a symbol.
- Add pseudo-localization and RTL test modes to development environments.
- Build translation extraction, catalog validation, and missing-translation checks into CI.

4. Failure behavior and operational expectations

- If a preferred locale is unavailable, fall back predictably and record a diagnostic rather than failing the entire request.
- If a time zone is invalid, use a documented tenant or product fallback and surface validation errors at configuration time.
- Never silently change data residency or legal-region behavior as a mere presentation fallback.

5. Suggested operational metrics

- Translation coverage and missing-key rate
- Locale formatting defects
- Requests by locale and region
- Timezone-related support incidents
- Regional configuration validation failures

6. Readiness and acceptance checklist

- ■ A product can add a locale without changing domain business logic.
- ■ Dates, numbers, and currencies are rendered using explicit locale and time-zone context.
- ■ RTL and long translated strings pass automated UI checks.
- ■ Fallback behavior is documented and tested.
- ■ Regional data handling is enforced independently from display locale.

Ownership note: assign a named team, service owner, support route, versioning policy, and deprecation process before treating this capability as a platform contract.